

My target task

My target task is developing a set of learning objectives for a session in the Anesthesia and Surgery B course in the second year of our pre-clinical curriculum. The overall focus of the session is developing student abilities to read and interpret spay/neuter research and discuss it with clients and colleagues.

Version 1 (Zero-shot)

Prompt

Write a set of 3-4 learning objectives for a second-year pre-clinical session for veterinary students focused on developing student abilities to read and interpret spay/neuter research and discuss it with clients and colleagues.

Output quality: Rank on a 1-5 scale

3

Two weaknesses I identified

They're packed too full of tasks, and some of the objectives double up on verbs.

Version 2 (add role specification)

Prompt

You are an experienced instructional designer who specializes course design (including objectives, assessments, and materials) for professional graduate programs.

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Output quality: Rank on a 1-5 scale

4-4.5

What improved from V1

Divided them into clear, brief objectives which were then elucidated with component parts. Used better, more actionable verbs.

Version 3 (few-shot example)

Prompt

Here is an example of a high-quality set of learning objectives from the first session of a veterinary curriculum.

- Predict how derangements in the cell manifest clinically by integrating knowledge of normal cell structure and function.
- Predict how derangements in metabolism manifest clinically by integrating knowledge of normal protein, carbohydrate, and lipid metabolism.
- Appraise the reliability and usefulness of resources when applying basic science to clinical cases.

You are an experienced instructional designer who specializes course design (including objectives, assessments, and materials) for professional graduate programs.

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Output quality: Rank on a 1-5 scale

2

What improved from V2

Nothing improved. The quality got markedly worse. The first two objectives are now nonsensical because the model decided that the “predict” verbs used in the exemplar were a key feature that it needed to emulate. The third and fourth objectives are now also less targeted to the goal of the session. This was a huge step backward.

Version 4 (chain-of-thought instruction)

Prompt

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You are an experienced instructional designer who specializes course design (including objectives, assessments, and materials) for professional graduate programs.

Write a set of 3-4 learning objectives for a second-year pre-clinical session for veterinary students focused on developing student abilities to read and interpret spay/neuter research and discuss it with clients and colleagues.

Before writing, think step by step through: (1) Who is the audience? (2) What is the single most important point? (3) What tone is appropriate? Then produce the output.

Output quality: Rank on a 1-5 scale

2

What improved from V3

Nothing improved. Again, the model decided to mimic the exemplar set of objectives much too closely and in ways that just don't make sense. It actually dropped the objective from V3 that was the best match for the purpose of the session. Somehow a step even further back.

Reflection

Adding a role clearly produced the biggest jump, although that's only because it was the only step that actually improved the output. V3 and V4 were both steps backward, and V3 was a huge step backward. My takeaway from this is that you need to be very careful with the exemplars that you give an LLM because it may not understand what aspects of the exemplar it should emulate.

What remains unresolved (or more accurately, what I need to work on more) is how to get the model to appropriately consider an example/exemplar without trying to emulate it in ways that deviate from the primary purpose it's been given. It would probably deal much better with a set of objectives from an entire module or an entire course than just the set from one session that I gave from it - too small of a sample size.

Appendix

Version 1 Output

1. **Critically evaluate spay/neuter research** by identifying study design (RCT, observational, retrospective), sample size, potential biases, and limitations; and distinguish between correlation and causation when interpreting results presented in veterinary literature.
2. **Translate complex research findings into clear, evidence-based messaging** appropriate for different audiences (pet owners, veterinary colleagues) by explaining key outcomes, uncertainty, and clinical applicability in accessible language while acknowledging areas of scientific debate.
3. **Compare and contrast contradictory evidence** on spay/neuter outcomes (e.g., reproductive cancers vs. orthopedic/behavioral conditions) and articulate how individual patient factors, breed, age, and local epidemiology inform personalized clinical recommendations.
4. **Engage in evidence-based dialogue** with clients and colleagues by confidently presenting research summaries, addressing common misconceptions, acknowledging knowledge gaps, and supporting individualized decision-making rather than one-size-fits-all protocols.

Version 2 Output

1. **Critically appraise spay/neuter research literature** by systematically evaluating study design, methodology, sample characteristics, and statistical reporting; identifying sources of bias; and assessing the strength of evidence using established frameworks (e.g., GRADE, levels of evidence hierarchies).
2. **Synthesize multi-source evidence on spay/neuter outcomes** by integrating findings across contradictory studies to construct nuanced, patient-centered clinical recommendations that account for breed predispositions, age at sterilization, individual risk factors, and gaps in current knowledge.
3. **Communicate research findings effectively across professional contexts** by translating research evidence into clear explanations for diverse audiences (pet owners, veterinary colleagues, specialists), distinguishing between established facts and areas of uncertainty, and justifying recommendations with appropriate confidence levels.
4. **Apply evidence-based reasoning to clinical decision-making conversations** by facilitating informed client/colleague discussions that acknowledge trade-offs between competing health outcomes, recognize legitimate differences in

professional opinion, and empower stakeholders to make decisions aligned with their values and risk tolerance.

Version 3 Output

1. **Predict how study design, bias, and statistical methods influence the reliability of spay/neuter research** by integrating knowledge of epidemiological principles, research methodology, and evidence hierarchies to assess which findings warrant clinical confidence.
2. **Predict how individual patient factors (breed, age, sex, health status, lifestyle) interact with spay/neuter outcomes** by integrating knowledge of reproductive endocrinology, cancer biology, orthopedic development, and behavioral physiology to contextualize population-level research findings.
3. **Appraise the reliability and usefulness of spay/neuter research resources** when communicating with clients and colleagues by critically evaluating study quality, recognizing the distinction between correlation and causation, and identifying when evidence is insufficient to support clinical recommendations.
4. **Construct evidence-informed spay/neuter recommendations** by integrating contradictory research findings, acknowledging areas of professional disagreement, and weighing competing health outcomes to facilitate informed decision-making that aligns with individual patient circumstances and client values.

Version 4 Output

Instructional Design Analysis

Step-by-Step Thinking:

(1) Who is the audience?

Second-year pre-clinical veterinary students—learners who have foundational science knowledge but are still developing clinical reasoning and professional identity. They need objectives that bridge basic science/research literacy with emerging clinical judgment.

(2) What is the single most important point?

The exemplar objectives reveal a curriculum philosophy: students should develop **predictive reasoning grounded in foundational knowledge**, not protocol memorization. The spay/neuter session should teach students to reason *through* research uncertainty rather than extract definitive answers *from* it.

(3) What tone is appropriate?

Intellectually rigorous but humble about complexity. The exemplar uses precise verbs (predict, appraise, integrate) and acknowledges that foundational knowledge enables

clinical application—not that research provides ready-made answers. The tone should reflect that evidence-based practice requires both critical thinking and comfort with ambiguity.

Learning Objectives: Reading and Interpreting Spay/Neuter Research

By the end of this session, students will be able to:

1. **Predict how study design and methodology influence research conclusions about spay/neuter outcomes** by integrating knowledge of epidemiological principles, statistical inference, and sources of bias to assess the reliability of reported findings.
2. **Predict how reproductive endocrinology, cancer biology, and orthopedic development interact with population-level spay/neuter research** by integrating foundational knowledge to contextualize which findings apply to individual patients and which remain uncertain.
3. **Appraise the reliability and usefulness of spay/neuter research resources** when advising clients and colleagues by critically evaluating evidence quality, distinguishing established findings from professional opinion, and identifying knowledge gaps.